

INSY336 - Data Handling and Coding for Analytics

Kyunghee Lee

About me

- Lee, Kyung + Hee
- Assistant Professor of Information Systems, McGill (2023–)
 - Wayne State University, Detroit (2017–2023)
 - Postdoctoral researcher, McGill (2016–2017)
- Research
 - How does digitization reshape firms and markets?
 - Technology adoption, digital transformation, and platform strategy

TA

- Jae Choi
- PhD student in Information Systems
- Your go-to for coding help, assignments, and grading

Should you still learn to code?

*"It is our job to create computing technology such that **nobody has to program**... Everybody in the world is now a programmer."*

Jensen Huang

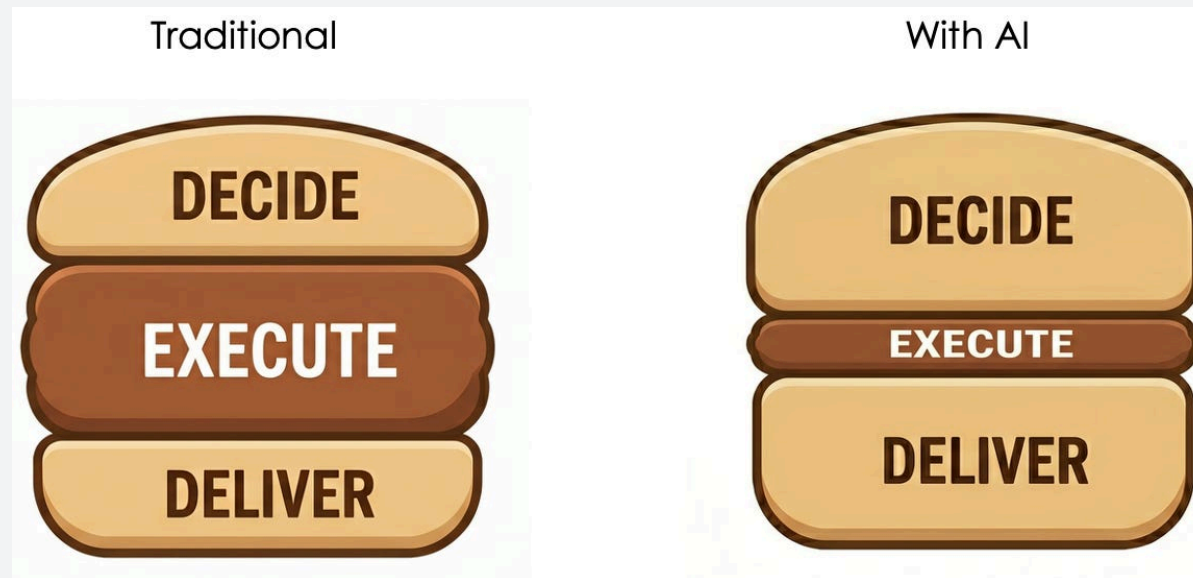
CEO, NVIDIA — 2024

*"[Discouraging people from learning programming] will be seen as some of the **worst career advice** ever given."*

Andrew Ng

Founder, Google Brain — 2025

A job is a bundle of tasks. AI kills some tasks, but not all.



AI agents write **8x** more code, but only **30%** more software ships.

Autor (2015, JEP); Acemoglu & Restrepo (2019, JEP); Narayanan & Kapoor (2026) · Score whole jobs: "47% at risk" → score tasks: 9% (OECD)

It's already in the data

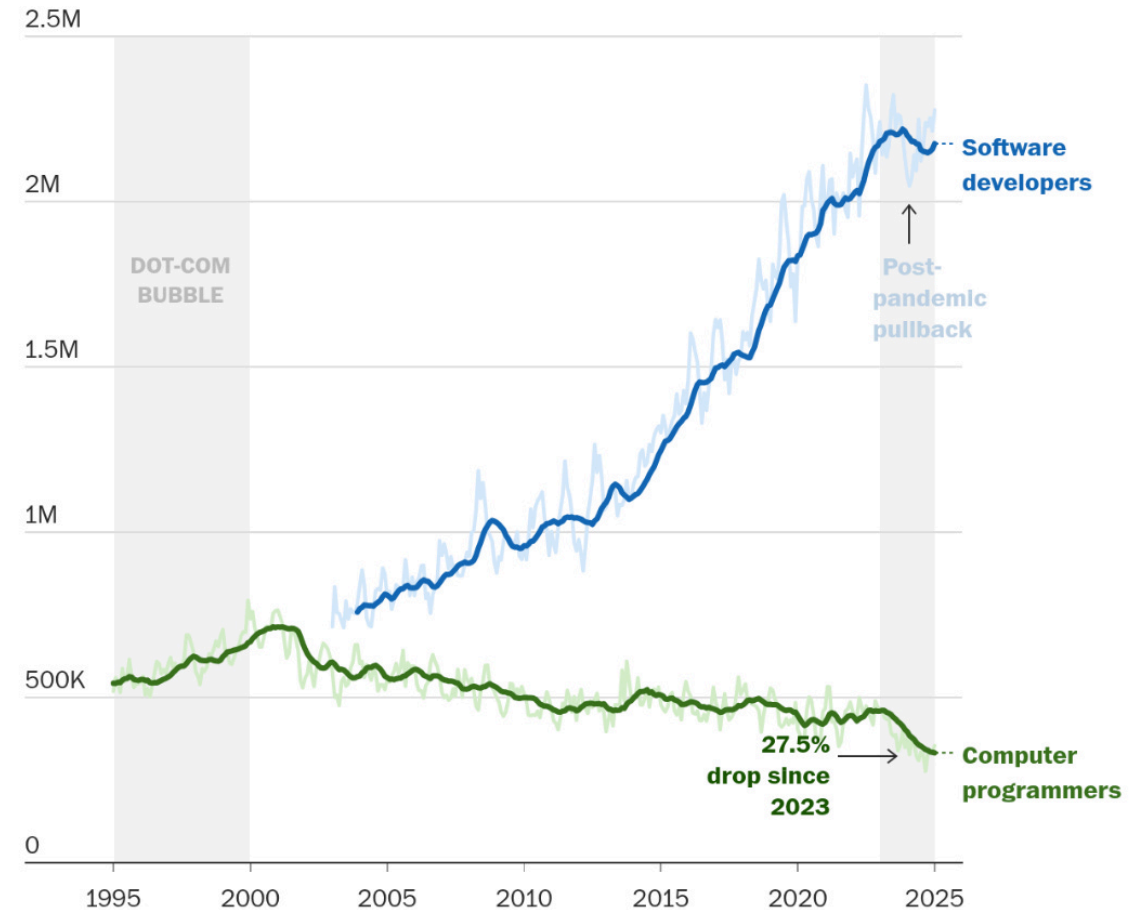
"Programmer" (the task): -27.5% since 2023.

"Software developer" (the job): near record highs.

Narayanan & Kapoor (2026)

Software developers vs Computer programmers

Over time, the industry has shifted toward employing broader developer roles that combine planning and coding, while traditional programming positions, which focus mainly on writing and testing the actual code, have declined.



Note: Dark line shows 12-month average. Data available for software developers starts in 2003.

Source: Current Population Survey from the Bureau of Labor Statistics via IPUMS.

WP INTELLIGENCE / THE WASHINGTON POST

We've seen this movie before

1979 — VisiCalc automates the spreadsheet. "Bookkeeping is done."

What actually happened in the U.S. after 1980:

- Bookkeeping clerks: **−400,000 jobs**
- Accountants and analysts: **+600,000 jobs**

The tool killed the ***task*** — and grew the ***field***.

NPR Planet Money Ep. 606: "Spreadsheets!"

Could this time be different?

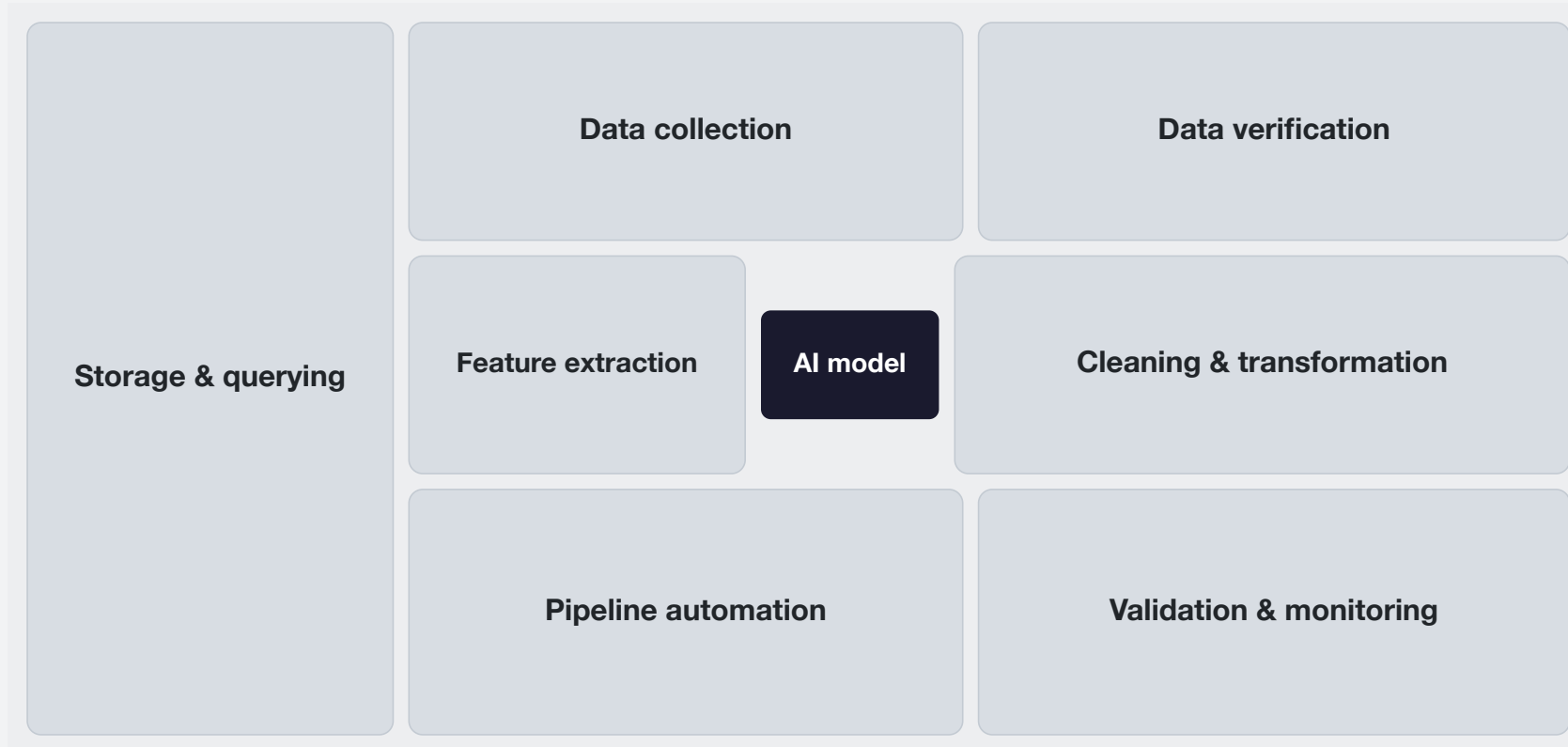
History repeats: AI follows the spreadsheet pattern — tasks die, the field grows.

This time is different: AI improves faster, and attacks many tasks at once.

The better question: what should we do?



Inside every AI project — it's not just chatting



The model is the smallest box — everything else is data work.

Data scientist, still the sexiest job?

Data Scientist: The Sexiest Job of the 21st Century

Meet the people who can coax treasure out of messy, unstructured data. by Thomas H. Davenport and DJ Patil

From the Magazine (October 2012)

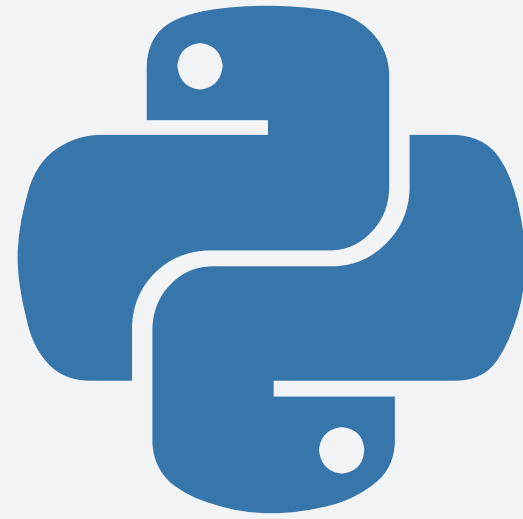
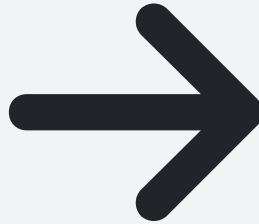
Is Data Scientist Still the Sexiest Job of the 21st Century?

by Thomas H. Davenport and DJ Patil

July 15, 2022

+33.5%

projected growth for **data scientists** by 2034
— the **4th fastest-growing** U.S. occupation



Course Objectives

- Manage and query data with SQL.
- Understand fundamental programming concepts in Python.
- Use Python to clean, transform, and explore data.
- Build an end-to-end **Extract-Transform-Load (ETL)** pipeline.

Module 1 - Data handling in SQL

- *Query and manage data in relational databases*

Module 2 - Fundamentals of programming in Python

- *Learn to code and think like a programmer*

Module 3 - Extract-Transform-Load (ETL)

- *Build an end-to-end data pipeline combining SQL + Python*

Languages we'll learn

SQL

```
SELECT * FROM table;
```

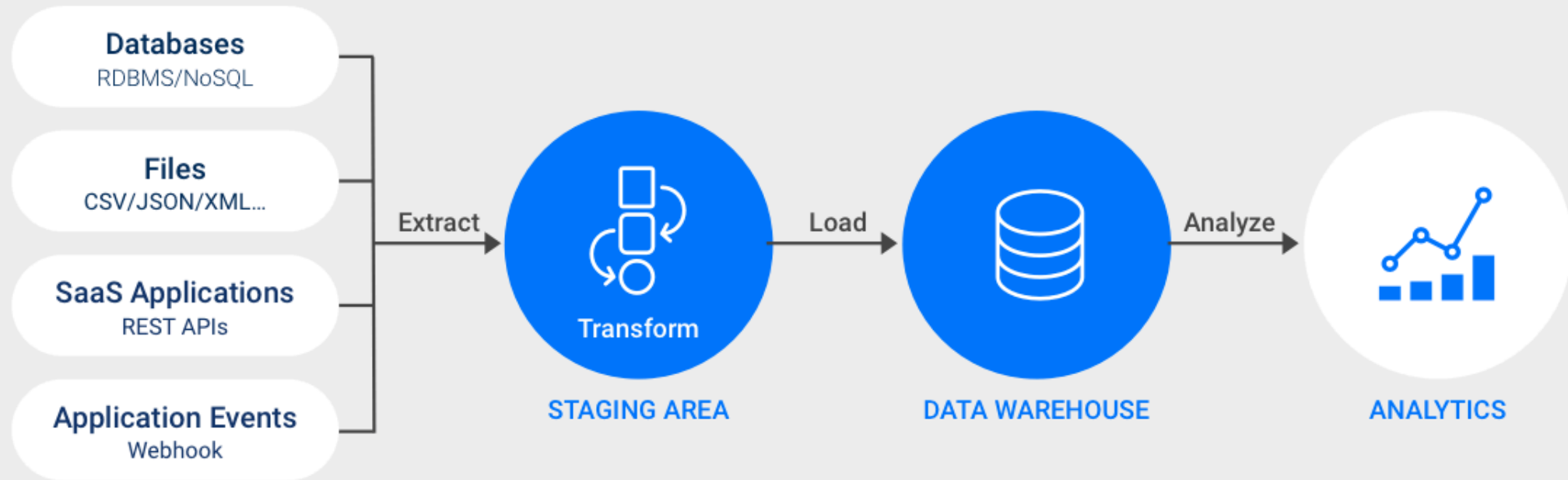
Python

```
def hello():  
    print("hello world")
```

REST API

```
import requests  
  
response = requests.get("https://api.example.com/data")  
data = response.json()
```

ETL PROCESS



Logistics

 **Course Home** — all readings, assignments, and resources

- Section 001: <https://insy336.github.io/f2026-001>
- Section 002: <https://insy336.github.io/f2026-002>

 **EdStem** — coding exercises, assignments, Q&A

 **MyCourses** — grades only

? Questions

- Content → post on EdStem discussion board
- Personal matters → email me or TA

Course Tools

- **EdStem**: browser-based coding platform
- **DataCamp**: take-home practice exercises
- **Tutorial session led by TA**: dates announced on EdStem

Coursework

- **Labs (in-class)**: every class, graded on completion — submit before class ends
- **Take-Home Exercises (4)**: DataCamp, multiple attempts, pass/fail
- **Assignments (3)**: build toward a full ETL pipeline, step by step
- **Quizzes (2)**: SQL & Python
- **Final Exam**: cumulative, bring it all together

Grading policy

Component	Weight (%)
Participation	2
Take-Home Exercise	5
Lab	8
Assignment	25
Quiz	30
Final Exam	30

- Late submission: **10%** deduction per day
- See **syllabus** for details

Getting to know each other

Team up and share:

-  Your background / major
-  Your career goals
-  What got you interested in this course
-  What you do for fun outside coding
-  Tips for surviving Montreal winter
-  The weirdest thing you've ever asked ChatGPT.
- **Which side are you on: Huang vs. Ng?**